

Multiresolution Ensemble Graph Learning for Phase Identification in Power Distribution Systems

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Abstract

Graphs have become a powerful tool in studying the underlying structure and behavior of complex networks and the analysis of data. Time series analysis is one such area where construction of graphs plays a critical role as graphs are helpful in studying the similarities or dissimilarities in time series data. Popular methods for constructing graphs on time series data include Similarity, CoAssociation Matrix, and Smoothness. While these methods rely on the variation of data in time (i.e. time difference), very less emphasis is put on volatility. Time series data such as those originating from smart meter data in power grids have a tendency to be volatile where a sudden spike or a tiny time shift may end up in different clustering results. We, therefore, propose building graphs for time-series based on multiresolution analysis that decomposes a signal using the Discrete Wavelet Transform (DWT). We then investigate the inference of underlying graph structures in each band as contributors to a final graph based clustering. This method is called multiview graph fusion where each wavelet resolution is treated as a view.

Method for Inferring Graphs from Data

The method given data $\mathbf{X} \in \mathbb{R}^{N \times M}$ each column is treated as a graph signal and a graph Laplacian is learnt based on the smoothness/total variation of the graph signals

$$\begin{aligned} \min_L \quad & \alpha \text{tr}(\mathbf{X}^T \mathbf{L} \mathbf{X}) + \beta \|\mathbf{L}\|_F^2 \\ \text{s.t.} \quad & \text{tr}(\mathbf{L}) = n, \\ & L_{ij} = L_{ji} \leq 0, i \neq j, \\ & \mathbf{L} \cdot \mathbf{1} = 0. \end{aligned} \quad (1)$$

where $\alpha \text{tr}(\mathbf{X}^T \mathbf{L} \mathbf{X})$ is the spatial smoothness term of graph signals and $\beta \|\mathbf{L}\|_F^2$ is the term for controlling the off-diagonal elements. α and β are called the regularization parameters.

Spectral Perturbation from Multiple Views

The graph laplacians are fused using the method of spectral perturbation and the resultant $\mathbf{L}^*$ which is given by the equation.

$$\mathbf{L}^* = \frac{\sum_{k=1}^K w^{(k)} \mathbf{U}_c^{(k)} \mathbf{\Lambda}_c^{(k)} (\mathbf{U}_c^{(k)})' + \alpha \sum_{k=1}^K w^{(k)} \mathbf{L}^{(k)}}{\sum_{k=1}^m w^{(k)} \mathbf{U}_c^{(k)} (\mathbf{U}_c^{(k)})' + \alpha \mathbf{I}} \quad (2)$$

where the $\mathbf{L}^k$ is the laplacian of the k -th band/view, $\mathbf{U}_c$ are the c largest eigenvectors and w^k are the weights assigned to each view/band. The values used for assigning weights to each multiresolution band in the final consensus matrix are derived from Similarity Induced Weights and modularity. Similarity induced weights are derived based on the similarity score between the pair of i and j views $sv_{ij} = sv_{ji} = \frac{\langle \mathbf{W}^{(i)}, \mathbf{W}^{(j)} \rangle_F}{\|\mathbf{W}^{(i)}\|_F \|\mathbf{W}^{(j)}\|_F}$. Where $\mathbf{W}$ are the weight matrices from the Laplacians. The modularity can be defined as for a graph $G(V, E)$ with an adjacency matrix $\mathbf{A}$;

$$\Sigma_k (e_{kk} - a_k^2) \quad (3)$$

where e_{kk} are the edges inside a cluster, S_k , while a_k is the average number of edges within S_k .

Conclusion and Future Work

In this work, we investigated a method inferring a graph based on multi-resolution analysis of time series signals and learning a final ensemble graph fused in a weighted fashion. The final procured graph has been applied for spectral clustering addressing the phase identification problem in power distribution systems. We would further investigate in applying this method for other time series clustering problems.

Time Series Data

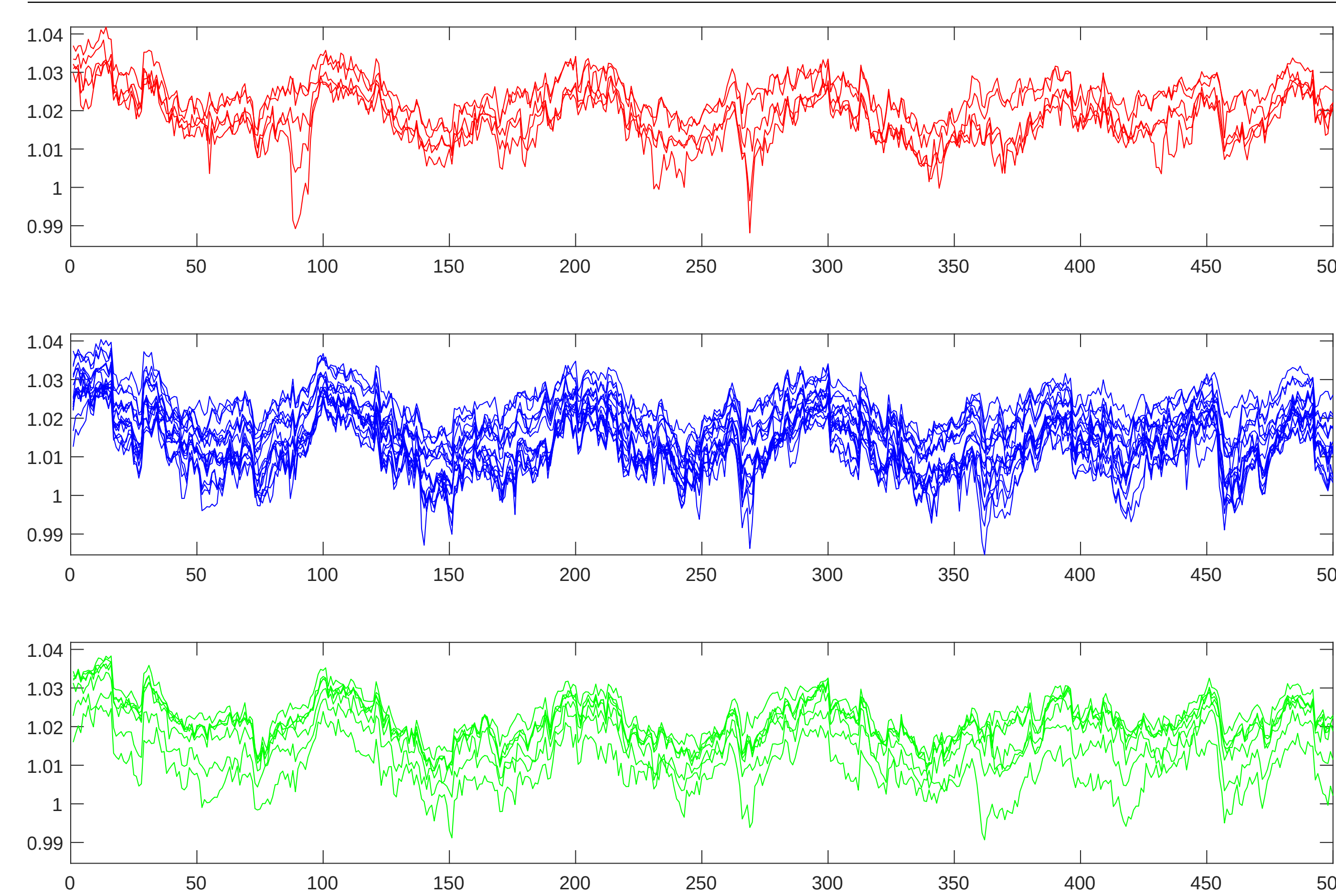


Figure 1. Historic Time Series Smart Meter Data

Example of Wavelet Decomposition of Time Series

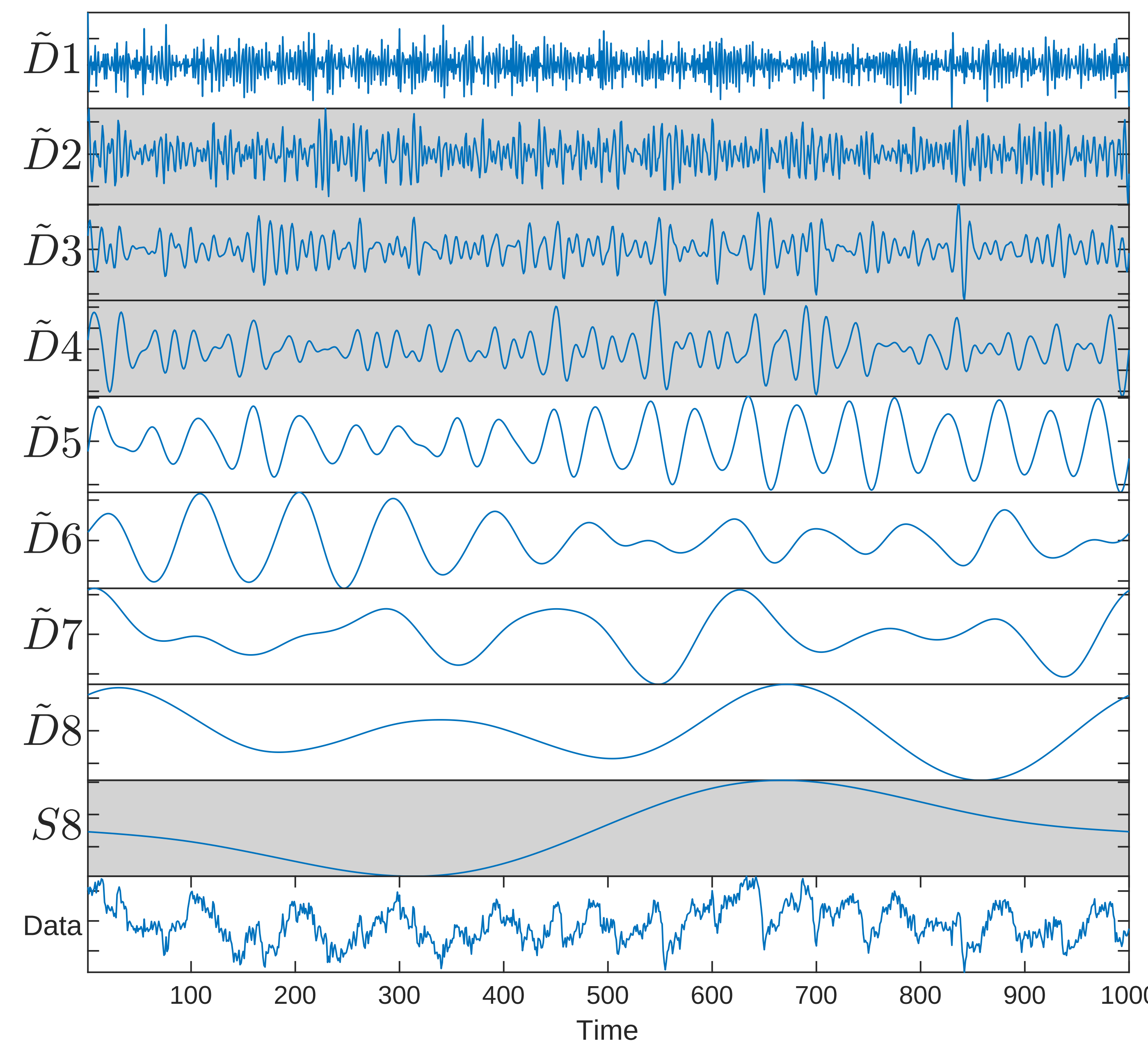


Figure 2. Wavelet Multi-resolution analysis of a time series signal

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Pipeline Multiview Consensus Graph Ensemble

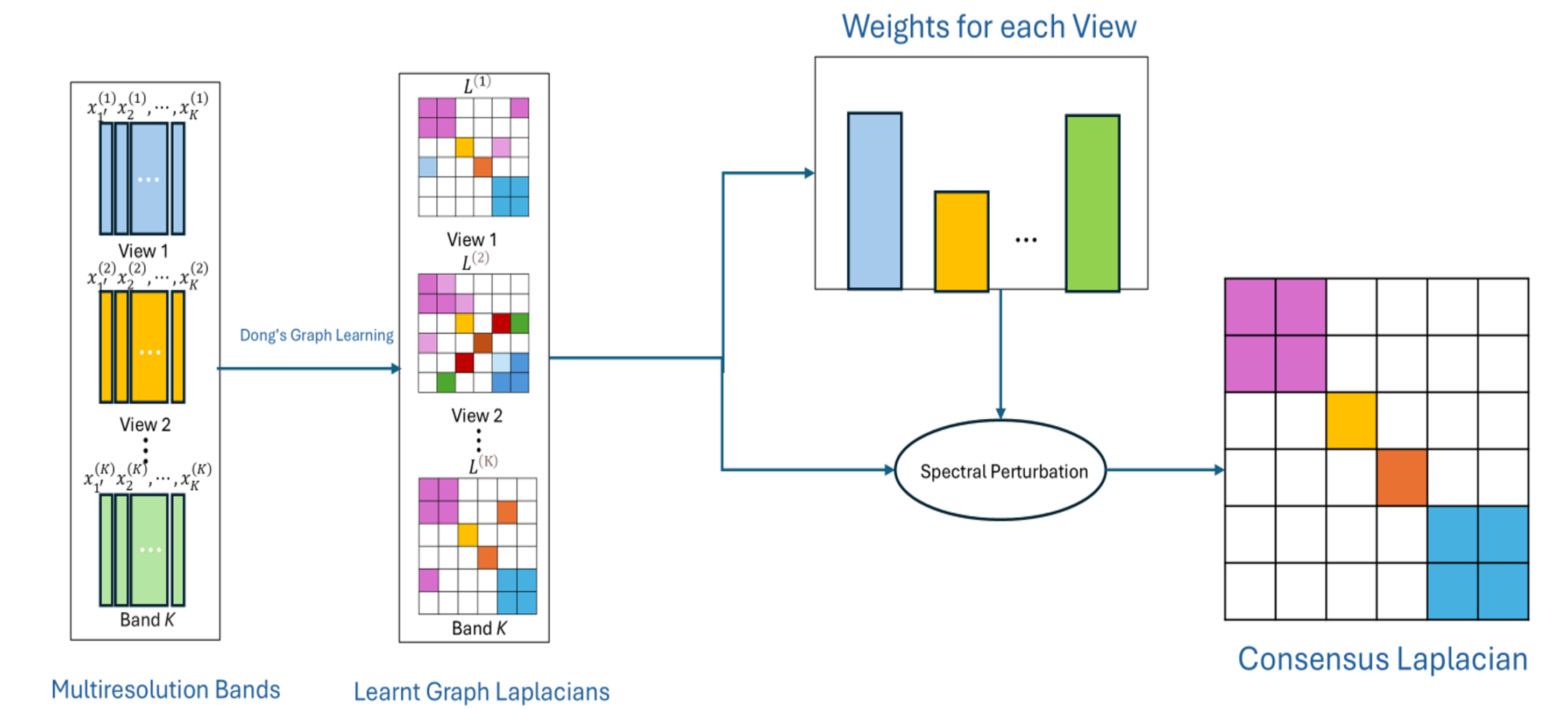


Figure 3. Pipeline for the Ensemble

Experimentation and Result

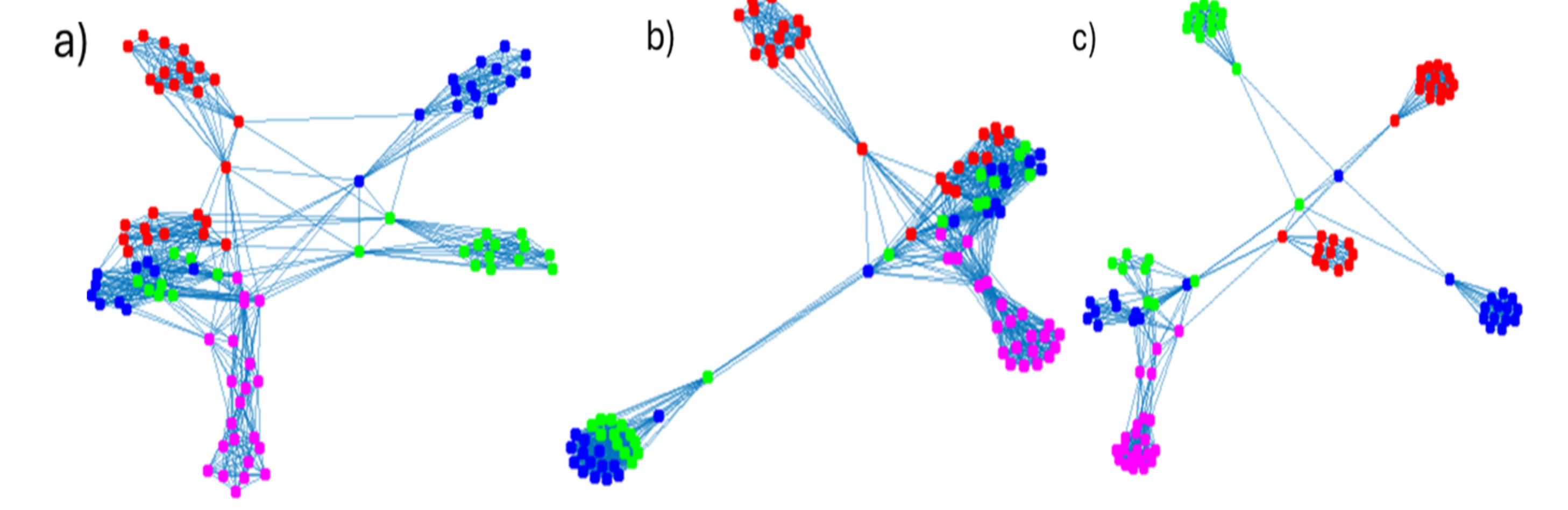


Figure 4. a) Graphs with k-nearest neighbors using similarity induced weights b) Graphs using Dong's algorithm and similarity induced weights c) Graphs using Dong's algorithm and modularity weights

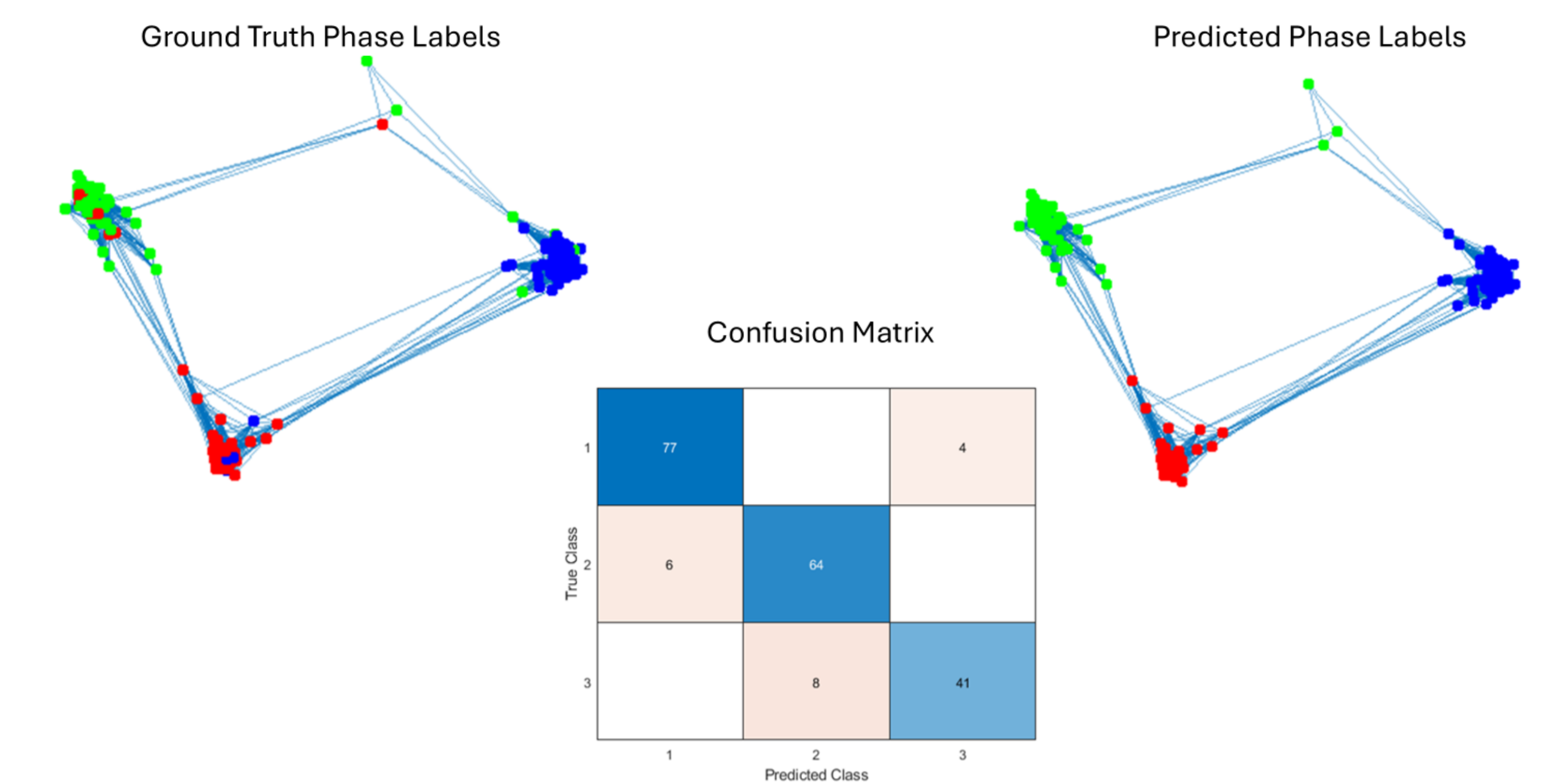


Figure 5. Phase Identification in Historic Smart Meter Data showing 90 percent accuracy